

Phishing Attack Simulation & Detection

Phishing:

Phishing is one of the most common social engineering attack techniques used by attackers to trick users into revealing sensitive information such as usernames, passwords, or financial details.

Objective:

The main objectives of this phishing simulation are:

- To understand phishing attack methodology
- To design a fake phishing email template
- To configure a landing page for credential harvesting
- To launch a phishing campaign using GoPhish
- To track user interaction and responses
- To identify phishing indicators (red flags)
- To learn prevention and mitigation techniques

Tools:

Kali Linux - Attacker Machine

Windows XP – Victim Machine

GoPhish -Phishing simulation framework

Lab Setup:

The phishing simulation was conducted in a controlled virtual lab environment using VMware.

- **Attacker Machine:** Kali Linux
 - IP Address: 192.168.136.128
- **Victim Machine:** Windows XP
 - IP Address: 192.168.136.129
- **Network Type:** VMware Host-Only Network

Both systems were connected to the same internal network to ensure isolation from the internet and external systems. Network connectivity between attacker and victim machines was verified using ICMP ping requests.

5. Phishing Attack Workflow

Phishing attacks generally follow a structured workflow where an attacker designs a deceptive message, delivers it to the target, and waits for user interaction. In this lab, the phishing attack lifecycle was simulated using GoPhish.

The workflow followed in this simulation is:

1. Creation of a phishing email template
2. Configuration of a phishing landing page
3. Setup of a sending profile
4. Launching the phishing campaign
5. Monitoring user interaction and results

This workflow helps in understanding how attackers operate and how phishing attempts are tracked.

GoPhish Configuration

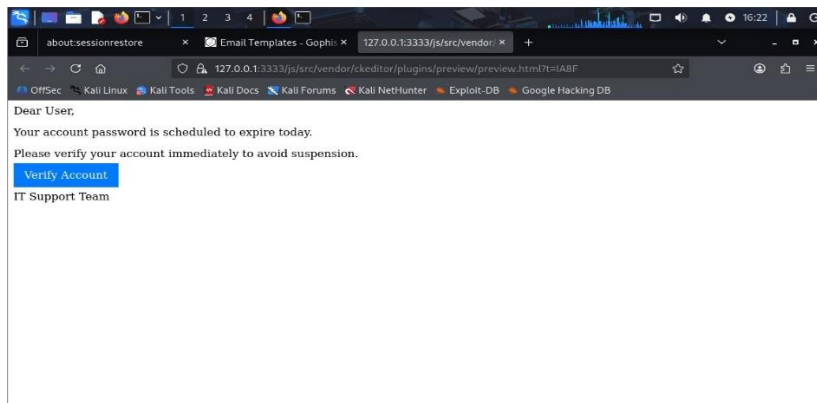
Email Template Creation:

A phishing email template was created to impersonate an internal IT security notification. The email was designed to create a sense of urgency and encourage the victim to click on the provided link.

Key characteristics of the email template:

- Professional and formal language
- Urgent subject line requesting immediate action
- Embedded hyperlink redirecting to a phishing landing page
- Spoofed sender name resembling an IT department

The purpose of this template was to simulate a realistic phishing email commonly seen in organizational environments.



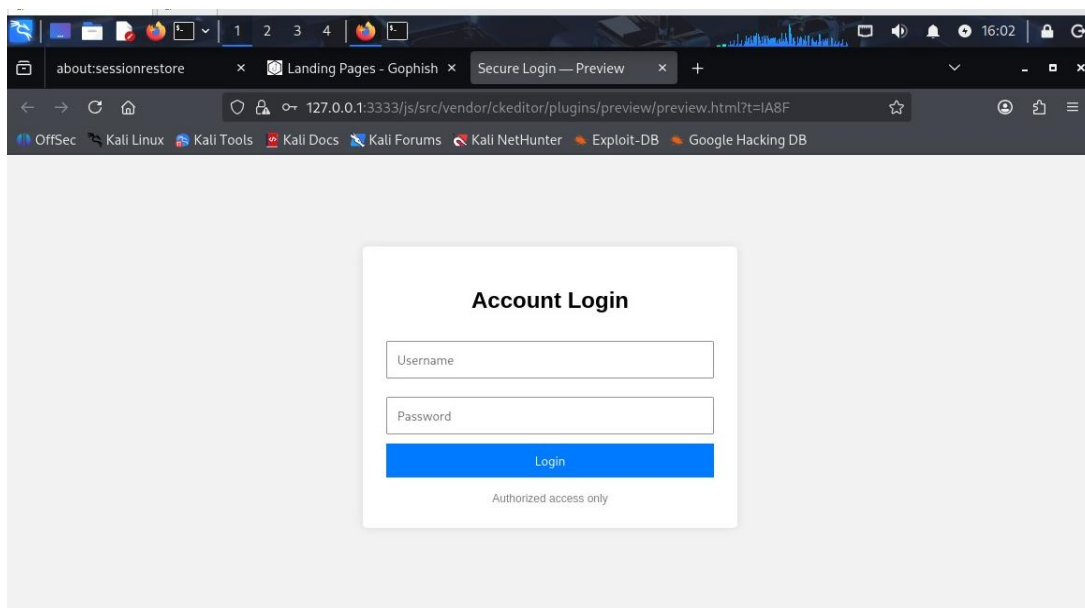
6.2 Landing Page Configuration

A fake login landing page was created to collect user credentials. The page visually resembled a legitimate internal login portal.

Landing page features:

- Username input field
- Password input field
- Submit button
- Credential capture enabled
- Redirect option after submission

When the victim enters credentials and submits the form, GoPhish records the data for analysis.



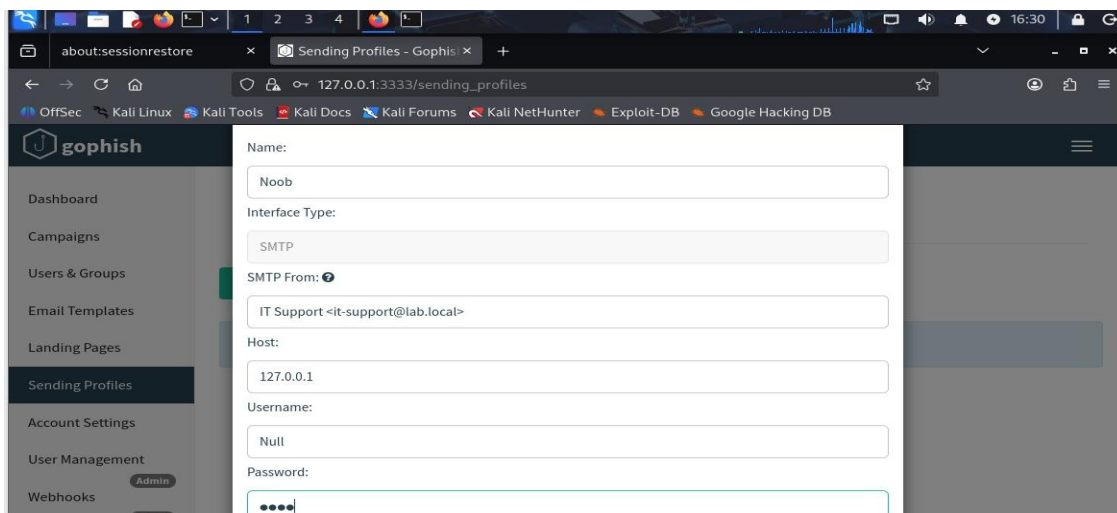
Sending Profile Configuration:

A sending profile was configured to define how phishing emails are sent. Due to the isolated lab environment, a **dummy SMTP configuration** was used instead of a real email server.

Sending profile details:

- SMTP Host: Dummy / Local configuration
- Username: Null
- Password: Null
- SMTP From Address: IT Support <it-support@lab.local>

⚠ Note: Real email delivery was not configured to avoid sending phishing emails over the internet.



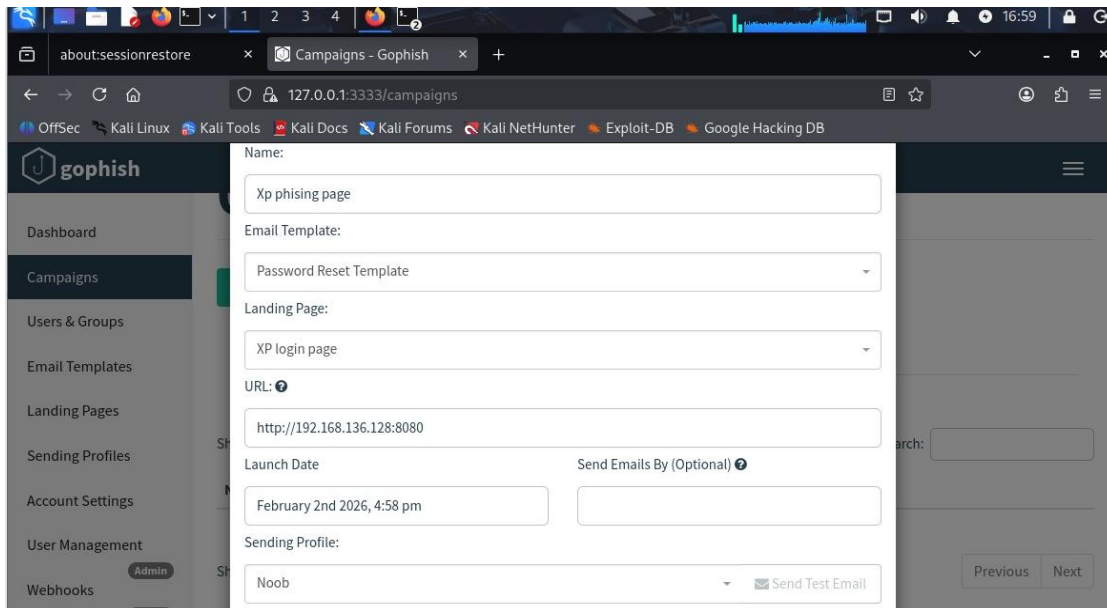
Campaign Creation:

A phishing campaign was created using the previously configured components.

Campaign components used:

- Email template
- Landing page
- Sending profile
- Target user profile

The campaign was launched to simulate a phishing attack against a single victim system in the lab.



Campaign Execution and Results:

Email Delivery Status:

When the campaign was launched, email delivery resulted in an error due to the dummy SMTP configuration. This behavior is expected in a lab environment without a real mail server. Despite the delivery failure, the campaign configuration and tracking logic remained functional.

User Interaction Simulation:

To simulate victim behavior, the phishing link generated by GoPhish was manually accessed from the Windows XP victim machine. This step simulates a scenario where a user clicks a phishing email link.

Credential Submission:

Dummy credentials were entered into the phishing landing page and submitted. GoPhish successfully captured and recorded the submitted credentials, demonstrating how phishing attacks harvest sensitive information.

Campaign Tracking and Monitoring:

GoPhish tracked the following actions:

- Phishing link access
- Credential submission

- Campaign activity status

These results demonstrate the effectiveness of phishing tracking mechanisms provided by GoPhish.

Red Flags Identified:

The phishing simulation highlights several indicators that users should watch for:

- Urgent or threatening email language
- Suspicious sender email address
- Requests for immediate credential verification
- Unexpected login requests
- Embedded hyperlinks redirecting to external pages

Awareness of these red flags helps reduce phishing success rates.

Prevention and Mitigation Techniques:

The following preventive measures can reduce phishing risks:

- Security awareness training
- Use of multi-factor authentication (MFA)
- Email filtering and spam detection
- Verification of email senders
- Avoid clicking unknown or suspicious links
- Reporting suspected phishing emails

Limitations:

- Real email delivery was not demonstrated due to the absence of a valid SMTP server.
- The phishing campaign was conducted in an isolated lab environment.
- User interaction was simulated manually.